



UNIVERSITY OF SAN-JOSE RECOLETOS

Engineering Mathematical Society

A.Y. 2026 - 2027



Prepared by:

EMS Academic Affairs Committee & Hon. Sophia Ragaza

“FUNCTIONS”

Practice Problems

Given: $g(x) = 3x^4 - 5x^2 + 2x - 7$

Find:

1. $g(2)$
2. $g(-2)$
3. $g(\frac{1}{2})$
4. $g(-x)$
5. $g(3)$

Given: $f(x) = 2x^2 - 3x + 4$

Find:

6. $f(x^2 - 3)$
7. $f(2x^2)$
8. $f(h + 1)$
9. $f(x + h)$
10. $f(3x - 2)$



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
EMS Academic Affairs Committee & Hon. Sophia Ragaza

“FUNCTIONS”
Step-By-Step Solutions

1. **Given:** $g(x) = 3x^4 - 5x^2 + 2x - 7$, find: $g(2)$

- **Rule:** Substitute the given value of x into the function.

- **Step 1:** Substitute $x = 2$ into the function.

$$\text{➤ } g(2) = 3(2)^4 - 5(2)^2 + 2(2) - 7$$

- **Step 2:** Evaluate the powers and multiplication.

$$\text{➤ } g(2) = 3(16) - 5(4) + 4 - 7$$

- **Step 3:** Simplify.

$$\text{➤ } g(2) = 48 - 20 + 4 - 7$$

Answer: $g(2) = 25$

2. **Given:** $g(x) = 3x^4 - 5x^2 + 2x - 7$, find: $g(-2)$

- **Rule:** Substitute the given value of x into the function.

- **Step 1:** Substitute $x = -2$ into the function.

$$\text{➤ } g(-2) = 3(-2)^4 - 5(-2)^2 + 2(-2) - 7$$

- **Step 2:** Evaluate the powers and multiplication.

$$\text{➤ } g(-2) = 3(16) - 5(4) - 4 - 7$$

- **Step 3:** Simplify.

$$\text{➤ } g(-2) = 48 - 20 - 4 - 7$$

Answer: $g(-2) = 17$



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
EMS Academic Affairs Committee & Hon. Sophia Ragaza

3. **Given:** $g(x) = 3x^4 - 5x^2 + 2x - 7$, find: $g(\frac{1}{2})$

- **Rule:** Substitute the given value of x into the function.

- **Step 1:** Substitute $x = \frac{1}{2}$ into the function.

$$\text{➤ } g(\frac{1}{2}) = 3(\frac{1}{2})^4 - 5(\frac{1}{2})^2 + 2(\frac{1}{2}) - 7$$

- **Step 2:** Evaluate the powers and multiplication.

$$\text{➤ } g(\frac{1}{2}) = 3(\frac{1}{16}) - 5(\frac{1}{4}) + 2(\frac{1}{2}) - 7$$

- **Step 3:** Simplify.

$$\text{➤ } g(\frac{1}{2}) = \frac{3}{16} - \frac{5}{4} + 1 - 7$$

$$\text{Answer: } g(\frac{1}{2}) = -\frac{113}{16}$$

4. **Given:** $g(x) = 3x^4 - 5x^2 + 2x - 7$, find: $g(-x)$

- **Rule:** Substitute the given value of x into the function.

- **Step 1:** Substitute $x = -x$ into the function.

$$\text{➤ } g(-x) = 3(-x)^4 - 5(-x)^2 + 2(-x) - 7$$

- **Step 2:** Evaluate the powers and multiplication.

$$\text{➤ } g(-x) = 3(x^4) - 5(x^2) + 2(-x) - 7$$

- **Step 3:** Check for like terms.

$$\text{➤ } g(-x) = 3x^4 - 5x^2 - 2x - 7$$

$$\text{Answer: } g(-x) = 3x^4 - 5x^2 - 2x - 7$$



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
EMS Academic Affairs Committee & Hon. Sophia Ragaza

5. **Given:** $g(x) = 3x^4 - 5x^2 + 2x - 7$, find: $g(3)$

- **Rule:** Substitute the given value of x into the function.

- **Step 1:** Substitute $x = 3$ into the function.

$$\triangleright g(3) = 3(3)^4 - 5(3)^2 + 2(3) - 7$$

- **Step 2:** Evaluate the powers and multiplication.

$$\triangleright g(3) = 3(81) - 5(9) + 6 - 7$$

- **Step 3:** Simplify.

$$\triangleright g(3) = 243 - 45 + 6 - 7$$

Answer: $g(3) = 197$

6. **Given:** $f(x) = 2x^2 - 3x + 4$, find: $f(x^2 - 3)$

- **Rule:** Substitute the given value of x into the function.

- **Step 1:** Substitute $x = x^2 - 3$ into the function.

$$\triangleright f(x^2 - 3) = 2(x^2 - 3)^2 - 3(x^2 - 3) + 4$$

- **Step 2:** Expand the squared expression.

$$\triangleright f(x^2 - 3) = 2(x^4 - 6x^2 + 9) - 3x^2 + 9 + 4$$

- **Step 3:** Distribute and combine like terms.

$$\triangleright f(x^2 - 3) = 2x^4 - 12x^2 + 18 - 3x^2 + 9 + 4$$

Answer: $f(x^2 - 3) = 2x^4 - 15x^2 + 31$

7. **Given:** $f(x) = 2x^2 - 3x + 4$, find: $f(2x^2)$

- **Rule:** Substitute the given value of x into the function.

- **Step 1:** Substitute $x = 2x^2$ into the function.



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
EMS Academic Affairs Committee & Hon. Sophia Ragaza

$$\triangleright f(2x^2) = 2(2x^2)^2 - 3(2x^2) + 4$$

- **Step 2:** Expand the squared expression.

$$\triangleright f(2x^2) = 2(4x^4) - 6x^2 + 4$$

- **Step 3:** Simplify.

$$\triangleright f(2x^2) = 8x^4 - 6x^2 + 4$$

Answer: $f(2x^2) = 8x^4 - 6x^2 + 4$

8. **Given:** $f(x) = 2x^2 - 3x + 4$, find: $f(h + 1)$

- **Rule:** Substitute the given value of x into the function.

- **Step 1:** Substitute $x = h + 1$ into the function.

$$\triangleright f(h + 1) = 2(h + 1)^2 - 3(h + 1) + 4$$

- **Step 2:** Expand the squared expression.

$$\triangleright f(h + 1) = 2(h^2 + 2h + 1) - 3h - 3 + 4$$

- **Step 3:** Distribute and combine like terms.

$$\triangleright f(h + 1) = 2h^2 + 4h + 2 - 3h - 3 + 4$$

Answer: $f(h + 1) = 2h^2 + h + 3$

9. **Given:** $f(x) = 2x^2 - 3x + 4$, find: $f(x + h)$

- **Rule:** Substitute the given value of x into the function.

- **Step 1:** Substitute $x = x + h$ into the function.

$$\triangleright f(x + h) = 2(x + h)^2 - 3(x + h) + 4$$

- **Step 2:** Expand the squared expression.

$$\triangleright f(x + h) = 2(x^2 + 2xh + h^2) - 3x - 3h + 4$$



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
EMS Academic Affairs Committee & Hon. Sophia Ragaza

- **Step 3:** Distribute and combine like terms.

$$\triangleright f(x + h) = 2x^2 + 4xh + 2h^2 - 3x - 3h + 4$$

Answer: $f(x + h) = 2x^2 + 4xh + 2h^2 - 3x - 3h + 4$

10. **Given:** $f(x) = 2x^2 - 3x + 4$, find: $f(3x - 2)$

- **Rule:** Substitute the given value of x into the function.
- **Step 1:** Substitute $x = 3x - 2$ into the function.

$$\triangleright f(3x - 2) = 2(3x - 2)^2 - 3(3x - 2) + 4$$

- **Step 2:** Expand the squared expression.

$$\triangleright f(3x - 2) = 2(9x^2 - 12x + 4) - 9x + 6 + 4$$

- **Step 3:** Distribute and combine like terms.

$$\triangleright f(3x - 2) = 18x^2 - 24x + 8 - 9x + 6 + 4$$

Answer: $f(3x - 2) = 18x^2 - 33x + 18$